

Due on monday, November 21st, at the beginning of class.

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1. Unitary matrices and operators.

(a) [**2 pts.**] Look up and report the definition of a unitary matrix.

(b) [**1 pt.**] Show that the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is its own inverse.

Hint: no need to calculate an inverse explicitly; use the fact/definition that a matrix multiplied by its inverse gives the identity.

(c) [**5 pts.**] Show that the three Pauli matrices are unitary.

(Comment: We have seen previously that they are also hermitian. These matrices are both hermitian and unitary!)

(d) [**3 pts.**] Show that the two-dimensional vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ has the same norm as $\sigma_y \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, i.e., operating with σ_y keeps the norm unchanged.

(Note: This is a special case of the fact that a unitary operation keeps the norm unchanged.)

(e) [**4 pts.**] If $\hat{U}$ is a unitary operator, show that the state $|\psi\rangle = \hat{U}|\phi\rangle$ has the same norm as the state $|\phi\rangle$.

2. [**7 pts.**] Look up the properties of the Dirac delta function. If a is a positive real number, write down the values of the following integrals.

$$\begin{array}{llll} \text{(a)} \int_{-a}^a \delta(x) & \text{(b)} \int_a^{2a} \delta(x) & \text{(c)} \int_{-a}^a \delta(x)f(x) & \text{(d)} \int_a^{2a} \delta(x)f(x) \\ \text{(e)} \int_{-a}^a \delta(x-2a) & \text{(f)} \int_{-a}^a \delta(x-2a)f(x) & \text{(g)} \int_0^{3a} \delta(x-2a)f(x) \end{array}$$

Hint: plotting the integrand in the region within the limits of integration is often helpful.

3. We will consider a particle in one dimension subject to a negative (attractive) delta potential at $x = 0$:

$$V(x) = -\lambda\delta(x)$$

This system has a single bound state at some negative energy, $E < 0$. We will find the energy and wavefunction of this bound state.

- (a) [**5 pts.**] Consider the left half-line ($x < 0$). Write down the general solution of the Schroedinger equation in this region. Remember: we are considering $E < 0$.

Argue why one of the terms of the general solution can be dropped.

- (b) [**2 pts.**] Similarly find the general solution on the right half-line ($x > 0$), and identify the term that should be dropped.

- (c) [**3 pts.**] Explain why the derivative of the wavefunction, $\psi'(x)$, need not be continuous at $x = 0$.

- (d) [**6 pts.**] Let ϵ be an infinitesimally small positive number. By integrating both sides of the Schroedinger equation from $-\epsilon$ to ϵ , show that the derivative of the wavefunction has the following discontinuity around the $x = 0$ point:

$$\psi'(\epsilon) - \psi'(-\epsilon) = -\frac{2m\lambda}{\hbar^2}\psi(0) + \text{infinitesimal term}$$

- (e) [**8 pts.**] Determine the energy of the bound state and the constants in the wavefunction.

You can use continuity of $\psi(x)$ and the normalization of the full wavefunction. In addition, the relation above for the discontinuity of $\psi'(x)$ can be used as a boundary condition:

$$\psi'_R(0) - \psi'_L(0) = -\frac{2m\lambda}{\hbar^2}\psi_L(0)$$

where $\psi_L(x)$ and $\psi_R(x)$ are the wavefunctions on the left half-line and right half-line respectively.

- (f) [**4 pts.**] Plot the wavefunction as a function of position. Make sure the (dis)continuity at $x = 0$ is clearly visible.